

Part 1

Practice Problems #14 - #16

Column-Row Expansion & Block Matrix | LU Factorization | Permuted LU & Vector Space

Practice problems #14

1. Let

$$A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix}.$$

- (a) Compute AB using row-column rule.
- (b) Compute AB using column-row expansion.
- (c) Let A and B be decomposed as follows, where A_{11} is of size 2×2 . Compute AB using block matrices.

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

2. Suppose that A_{11} is an invertible matrix. Find matrices X and Y such that the product below has the form indicated. Also, compute B_{22} .

$$\begin{bmatrix} I & 0 & 0 \\ X & I & 0 \\ Y & 0 & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} \\ 0 & B_{22} \\ 0 & B_{32} \end{bmatrix}.$$

3. Suppose A_{11} is invertible. Find X and Y so that

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix}$$

where $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$. The matrix S is called the *Schur complement* of A_{11} . Likewise, if A_{22} is invertible, the matrix $A_{11} - A_{12}A_{22}^{-1}A_{21}$ is called the Schur complement of A_{22} . Such expression occur frequently in the theory of systems engineering, and elsewhere (don't need to memorize).

Note that this is a factorization of A into block diagonal and block triangular matrices.

Answers or Hints are on page 2.

#1: Do them! You should get same answer for all.

#2: $X = -A_{21}A_{11}^{-1}$, $Y = -A_{31}A_{11}^{-1}$, $B_{22} = A_{22} - A_{21}A_{11}^{-1}A_{12}$

#3: $X = A_{21}A_{11}^{-1}$, $Y = A_{11}^{-1}A_{12}$

Practice problems #15

1. Let $A = \begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$.

(a) Find an LU factorization of A .

(b) Use (a) to solve the equation $Ax = b$.

2. True or False. Justify your answer:

Suppose a matrix A is row reduced to an echelon form U . If $A = LU$ is an LU factorization, then A can be row reduced to U using only replacement operations.

(This is asking the *converse* of what we did in class. In class, we showed that *if* A is row reduced to U using only replacement operations, *then* $A = LU$.)

Answers or Hints on page 2.

$$\#1: L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 3/2 & -5 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & -2 & 4 \\ 0 & -2 & -1 \\ 0 & 0 & -6 \end{bmatrix}, y = \begin{bmatrix} 0 \\ -5 \\ -18 \end{bmatrix}, x = \begin{bmatrix} -5 \\ 1 \\ 3 \end{bmatrix}$$

#2: True. (Since L means a unit lower triangular matrix in the LU factorization, L can be reduced to I using only which row operation? So, $E_k \cdots E_1 L = I$ where E_i are what kind of elementary matrices? Then, $A = LU$ becomes $E_k \cdots E_1 A = E_k \cdots E_1 LU$.)

Practice problems #16

1. Suppose $PA = LU$ where P is a permutation matrix. Explain the steps to solve the equation $Ax = b$.
2. Let $A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$.
 - (a) Give a permutation matrix P so that PA is upper triangular.
 - (b) Give two permutation matrices P_1 and P_2 so that P_1AP_2 is lower triangular.
3. Find a 3×3 permutation matrix $P \neq I$ with $P^3 = I$.
4. True or False. Justify.
 - (a) Suppose V is a set of all 2×2 real matrices with *nonnegative* entries. With the sum and scalar multiplications defined as usual (that is, elementwise, as we know of), V is a vector space.
 - (b) Suppose V is a set of 2×2 symmetric matrices. With the sum and scalar multiplications defined as usual (that is, elementwise, as we know of), V is a vector space.

Answers are on the back.

#1: Premultiply P to the equation to get $PAx = Pb$ (that is, permute entries of b) which then becomes $LUx = b'$, ($b' = Pb$). Solve this equation as usual LU .

$$\#2(a): P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\#2(b): P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\#3: P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ or } P^T.$$

#4(a): False. Hint: closed under scalar multiplication? counter-example? Note that to be a vector space, the scalar multiplication thing should be true for *all* scalars.

#4(b): True. Sum of two symmetric matrices symmetric? How about scalar multiple? Is there zero vector? (Is zero matrix symmetric?) Is additive inverse of a symmetric matrix symmetric? Must show all (10) axioms are satisfied.

Part 2

Practice Problems #17

Subspace

Practice Problems #17

1. True or False. Justify your answer.

Let $V = \mathbf{R}^2$, which we showed is a vector space with the usual operations

$$u + v = (u_1 + v_1, u_2 + v_2), \quad cu = (cu_1, cu_2).$$

Then,

$$H = \{(x, y) : xy \leq 0\},$$

is a subspace of V .

2. Show $c0 = 0$ for every scalar c following the proof below. Justify each step.

Proof: $c0 = c(0 + 0) = \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$.

Then,

$$0 = -c0 + c0 = (-c0 + c0) + \underline{\hspace{2cm}} = 0 + c0 = \underline{\hspace{2cm}}.$$

3. For $u \in V$, where V is a vector space, suppose

$$\alpha u = 0$$

for some nonzero scalar α . Show that $u = 0$. Justify each and every step in your proof.

Hints or Answers are on the next page.

#1: False. Why?

#2: Mimic what we did in class.

#3: Note that $u = 1u$ (which axiom?). Use the result from #2.

Part 3

Practice Problems #18 - #26

Determinants | Determinant Properties | Cramer's Rule & Adjugate
Area, Volume & Linear Transformations | Col A & Ker T | Bases
Coordinate Mapping | Dimension | Change of Basis

Practice problems #18

1. We know that $\mathbf{R}^{2 \times 2}$ is a vector space with usual element-wise addition and scalar multiplication.

- (a) Show that the set H of all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ is a subspace of $\mathbf{R}^{2 \times 2}$.
- (b) Give a spanning set for H .

2. Compute the determinant of the matrix A using

- (a) cofactor expansion
- (b) short-cut formula given in class.

$$A = \begin{bmatrix} 1 & -2 & 7 \\ -4 & 8 & 5 \\ 2 & -4 & 3 \end{bmatrix}.$$

3. Compute the determinant.

$$A = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 2 & 7 & 0 & 6 \\ 0 & 6 & 3 & 0 \\ 7 & 3 & 1 & -5 \end{bmatrix}$$

Answers or Hints on page 2.

#1(a): Check 4 items as we did in class.

#1(b): $H = \text{span}$ of what *vectors*? (Here, vectors are matrices.) One *possible* answer is

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

#2: 0 for both

#3: -546

Practice problems #19

1. Find the determinant by row reduction to echelon form:

$$A = \begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{bmatrix}$$

2. Combine the methods of row reduction and cofactor expansion to compute the determinant:

$$A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}$$

3. Let A and B be 4×4 matrices, with $\det A = -1$ and $\det B = 2$. Compute:

- (a) $\det AB$
- (b) $\det (B^5)$
- (c) $\det 2A$

4. True or False. Justify your answer.

- (a) Let A and B be square matrices, with B invertible. Then, $\det(BAB^{-1}) = \det A$.
- (b) Suppose that A is a square matrix such that $\det(A^4) = 0$. Then, columns of A are linearly independent.
- (c) Let U be a square matrix such that $U^T U = I$. Then $\det U = \pm 1$.

Answers or Hints on page 2.

#1: 24

#2: 0

#3: (a) -2 , (b) 32 , (c) -16

#4: (a) T, (b) F (c) T. Justify!

Practice problems #20

1. Determine the values of the parameter s for which the given system has a unique solution, and then describe the solution using Cramer's Rule.

$$\begin{cases} sx_1 + 2sx_2 &= -1 \\ 3x_1 + 6sx_2 &= 4 \end{cases}$$

2. Compute the adjugate of the given matrix, and the inverse of the matrix in terms of adjugate.

$$\begin{bmatrix} 3 & 6 & 7 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}$$

3. True or False. Justify your answer.

Suppose that all the entries in A are integers and $\det A = 1$. Then, all the entries in A^{-1} are integers.

Answers or Hints on page 2.

$$\#1: s \neq 0, 1; \quad x_1 = \frac{-7}{3(s-1)}, \quad x_2 = \frac{4s+3}{6s(s-1)}.$$

$$\#2: \operatorname{adj} A = \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}, \quad A^{-1} = (-1) \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

#3: True. (Hint: True for 2×2 case? How about in..... proof?)

Practice problems #21

1. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1, 4, 0)$, $(-2, -5, 2)$, $(-1, 2, -1)$.
2. Let R be the triangle with vertices at (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Show that area of the triangle R equals $\frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|$.
3. Let $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be the linear transformation that maps e_1, e_2, e_3 to ae_1, be_2, ce_3 , respectively, where a, b and c are some positive real numbers. Let S be the unit ball in $\mathbf{R}^3$ bounded by the unit sphere $x^2 + y^2 + z^2 = 1$. Use the fact that the volume of the unit ball is $4\pi/3$ to find the volume of $T(S)$.

Answers or Hints are on page 2.

#1: 15

#2: The area of R is $\frac{1}{2} \left| \det \begin{bmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{bmatrix} \right|$, and the matrix $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ is row

equivalent to the matrix $\begin{bmatrix} x_1 - x_3 & y_1 - y_3 & 0 \\ x_2 - x_3 & y_2 - y_3 & 0 \\ x_3 & y_3 & 1 \end{bmatrix}$.

#3: $abc4\pi/3$

Practice problems #22

1. Find A such that the given set is $\text{Col } A$:

$$\left\{ \begin{bmatrix} b - c \\ 2b + c + d \\ 5c - 4d \\ d \end{bmatrix} : b, c, d \text{ real} \right\}.$$

2. Let $T : V \rightarrow W$ be a linear transformation from a vector space V into a vector space W . Prove that $\text{Range } T$ is a subspace of W .

3. Define a linear transformation $T : \mathbf{P}_2 \rightarrow \mathbf{R}^2$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$. For instance, if

$$p(t) = 3 + 5t + 7t^2, \text{ then } T(p) = \begin{bmatrix} 3 \\ 15 \end{bmatrix}.$$

- (a) Show that T is a linear transformation.
- (b) Find a polynomial $p \in \mathbf{P}_2$ that spans the Kernel of T , i.e., find a spanning set for $\text{Ker } T$.
- (c) Describe the range of T .

Answers are on page 2.

$$\#1: \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

#2: Note that $\text{Range } T = \{b \in W : T(x) = b \text{ for some } x \in V\}$.

$\text{Range } T \subseteq W$ is clear.

The zero vector of W is in range of T because $T(0) = 0$ (since T is linear).

If b_1, b_2 are in $\text{Range } T$ (which means there are x_1 and x_2 in V such that $T(x_1) = b_1$ and $T(x_2) = b_2$), then $b_1 + b_2 \in \text{Range } T$ because $T(x) = b_1 + b_2$ has a solution (what's the solution?)

Show similarly that if $b \in \text{Range } T$ and c is any scalar, then $T(x) = cb$ has a solution (what's the solution?) so that $cb \in \text{Range } T$.

#3(a): Let p, q be polynomials in $\mathbf{P}_2$, then

$$T(p+q) = \begin{bmatrix} (p+q)(0) \\ (p+q)(1) \end{bmatrix} = \begin{bmatrix} p(0) + q(0) \\ p(1) + q(1) \end{bmatrix} = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix} + \begin{bmatrix} q(0) \\ q(1) \end{bmatrix} = T(p) + T(q).$$

Similarly, show that $T(cp) = cT(p)$.

#3(b): $\text{Ker } T = \{p \in \mathbf{P}_2 : p(0) = 0 \text{ and } p(1) = 0\} = \{\alpha t(t-1) : \alpha \in \mathbf{R}\}$, i.e., $\text{Ker } T = \text{Span } \{t(t-1)\}$. So, the spanning set for $\text{Ker } T$ is $\{t(t-1)\}$.

#3(c): Given any vector $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \in \mathbf{R}^2$, we can form a polynomial p so that $p(0) = \alpha$ and $p(1) = \beta$. For instance, $p(t) = (\beta - \alpha)t + \alpha$ is such polynomial.
Hence $\text{Range } T = \mathbf{R}^2$.

Practice problems #23

1. True or False. Justify your answer:

(a) The set $B = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ in $\mathbf{P}_2$ is linearly independent.

(b) The set $B = \left\{ \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \begin{bmatrix} 0 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix} \right\}$ in $\mathbf{R}^{4 \times 5}$ is linearly independent.

2. Find a basis for the set of vectors (x, y, z) in $\mathbf{R}^3$ such that $x + 2y + z = 0$.

3. Find bases for $\text{Nul } A$ and $\text{Col } A$, where A is row equivalent to B :

$$A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 0 & 5 & -7 & 8 \\ 0 & 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Let $v_1 = \begin{bmatrix} 4 \\ -3 \\ 7 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 9 \\ -2 \end{bmatrix}$, $v_3 = \begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}$, and $H = \text{Span}\{v_1, v_2, v_3\}$. It can be verified that $4v_1 + 5v_2 - 3v_3 = 0$. Use this information to find a basis for H .

5. True or False. Justify:

(a) Suppose $v_1, \dots, v_4$ are vectors in $\mathbf{R}^4$ such that $\mathbf{R}^4 = \text{Span}\{v_1, \dots, v_4\}$. Then $\{v_1, \dots, v_4\}$ is a basis for $\mathbf{R}^4$.

(b) Let $B = \{v_1, \dots, v_n\}$ be a linearly independent set in $\mathbf{R}^n$. Then B is a basis for $\mathbf{R}^n$.

(c) Every vector space has a basis.

Answers are on the back.

Answers:

#1(a): True. Consider the equation $c_1(1 + t^2) + c_2(t + t^2) + c_3(1 + 2t + t^2) = 0$.

#1(b): True. Justify!

#2: This problem is essentially asking to find a basis for $\text{Nul } A$ for an appropriate matrix

A. One possible answer: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

#3: Basis for $\text{Nul } A$: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

Basis for $\text{Col } A$: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -5 \\ -5 \\ 0 \\ -5 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 5 \\ -2 \end{bmatrix} \right\}$

#4: $\{v_1, v_2\}$ or $\{v_1, v_3\}$ or $\{v_2, v_3\}$ are all possible answers.

#5(a): True. Let $A = [v_1 \cdots v_4]$. Then, A is a 4×4 matrix. If $\mathbf{R}^4 = \text{Span } \{v_1, \dots, v_4\}$, what can you say about the columns of A ?

#5(b): True. Note that the matrix $A = [v_1 \cdots v_n]$ is $n \times n$.

#5(c): False. Give an example of a vector space with no basis.

Practice problems #24

1. Let V be a vector space with a basis $B = \{b_1, b_2, \dots, b_n\}$. Show that the coordinate mapping $T : V \rightarrow \mathbf{R}^n$ defined by $T(x) = [x]_B$ satisfies

$$T(cx) = cT(x)$$

for all $x \in V$ and for all scalars c .

2. Find the vector x determined by the given coordinate vector $[x]_B$ and the given basis $B = \{b_1, b_2, b_3\}$.

$$b_1 = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, b_2 = \begin{bmatrix} 5 \\ 2 \\ -2 \end{bmatrix}, b_3 = \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix}, [x]_B = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}.$$

3. Prove that the set $B = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ is a basis for $\mathbf{P}_2$.

Answers or Hints are on page 2.

#1: Mimic what we did in class.

#2:
$$\begin{bmatrix} -1 \\ -5 \\ 9 \end{bmatrix}$$

#3: Show two things!

Practice Problems #25

1. In the following, V is a nonzero finite-dimensional vector space, and the vectors listed belong to V .

True or False. Justify:

- (a) If there exists a set $\{v_1, \dots, v_p\}$ that spans V , then $\dim V \leq p$.
 - (b) If there exists a linearly independent set $\{v_1, \dots, v_p\}$ in V , then $\dim V \geq p$.
 - (c) If $\dim V = p$, then there exists a spanning set of $p + 1$ vectors in V .
 - (d) If there exists a linearly dependent set $\{v_1, \dots, v_p\}$ in V , then $\dim V \leq p$.
2. Prove that the first four Hermite polynomials $1, 2t, -2 + 4t^2$, and $-12t + 8t^3$ are linearly independent in $\mathbf{P}_3$ using Coordinate mapping idea. Then, explain why the set $S = \{1, 2t, -2 + 4t^2, -12t + 8t^3\}$ must be a basis of $\mathbf{P}_3$.

Answers or Hints are on page 2.

#1(a) T, (b) T, (c) T (justify them!), (d) F (create a counterexample)

#2: Mimic what we did in class. What is $\dim \mathbf{P}_3$? How many vectors in S ?

Practice problems #26

1. In $\mathbf{P}_2$, find the change-of-coordinates matrix from the basis $B = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ to the standard basis $S = \{1, t, t^2\}$.
If $p(t) = -1 + 2t$, what is the S -coordinate vector for $p(t)$?
Then find the B -coordinate vector for $p(t)$ *using the change-of-coordinates matrix*.
2. In $\mathbf{P}_1$, the following B and C are two bases. Find the change-of-coordinates matrix from B to C , and the change-of-coordinates matrix from C to B .

$$B = \{7 + 5t, -3 - t\}, \quad C = \{1 - 5t, -2 + 2t\}$$

Hints or Answers are on page 2.

#1: $M_{S \leftarrow B} = \begin{bmatrix} [b_1]_S & [b_2]_S & [b_3]_S \end{bmatrix} = \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$. What is $[p(t)]_S$?

$$[p(t)]_B = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

#2: Using isomorphism idea from $\mathbf{P}_1$ onto $\mathbf{R}^2$, that is, by using coordinate mapping relative to the standard basis $S = \{1, t\}$, it would be better to write the bases B and C as set of vectors in $\mathbf{R}^2$. Then, you can apply the algorithm that we did in class to find the change of coordinate matrix $M_{C \leftarrow B}$.

Ans:

$$M_{C \leftarrow B} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}, \quad \text{and} \quad M_{B \leftarrow C} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}$$

Note that #1 is easier with definitions, #2 is easier with algorithms.

Part 4

Practice Problems #27

Row Space

Practice problems #27

1. Find bases for $\text{Row}A$ and $\text{Row}B$, where A is row equivalent to B :

$$A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 0 & 5 & -7 & 8 \\ 0 & 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

2. Suppose that a nonhomogeneous system of six linear equations in eight unknowns has a solution, with two free variables. Is it possible to change some constants on the equations' right-hand-sides to make the new system inconsistent? Explain.
3. Let A be an $m \times n$ matrix. Which of the subspaces $\text{Row } A$, $\text{Col } A$, $\text{Nul } A$, $\text{Row } A^T$, $\text{Col } A^T$, and $\text{Nul } A^T$ are in $\mathbf{R}^m$ and which are in $\mathbf{R}^n$? How many distinct subspaces are in this list?
4. Let A be an $m \times n$ matrix. Prove that the equation $Ax = b$ has a solution for all $b \in \mathbf{R}^m$ if and only if the equation $A^T x = 0$ has only the trivial solution.

Answers or Hints are on page 2.

#1: Easy. One answer works for both $\text{Row } A$ and $\text{Row } B$.

#2: No. (nullity = 2, so rank = 6, hence??)

#3: Subspaces of $\mathbf{R}^n$ are $\text{Row } A$ and $\text{Nul } A$; Subspaces of $\mathbf{R}^m$ are $\text{Col } A$ and $\text{Nul } A^T$; four distinct subspaces.

#4: Hint: $\text{rank } A^T = \dim \text{Col } A^T$ (by definition) = $\dim \text{Row } A$ (showed in class)